Business Question and Visualization Report

Date	28 October 2026
Team ID	EN25CS3010020
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

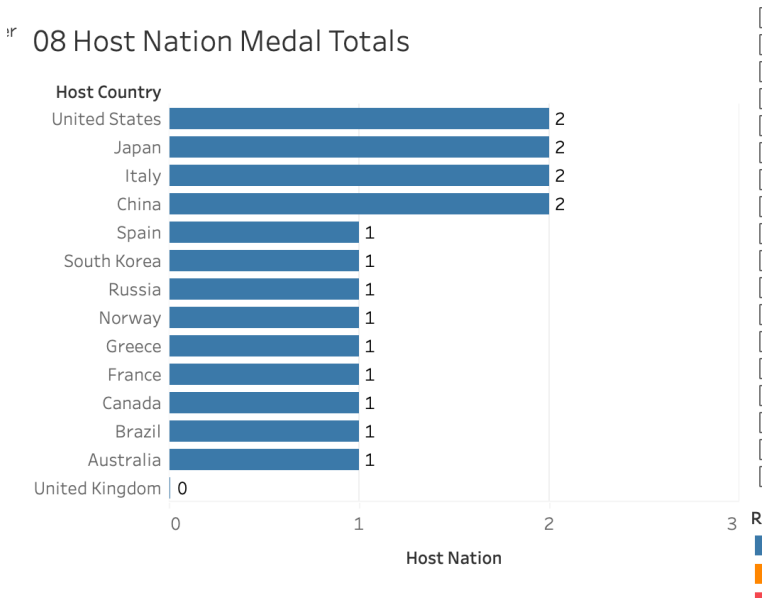
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

Sample

1. **Which countries have the highest total Olympic medal counts?**
   - ***Code***

```
SELECT
   Country,
   SUM(Total_Medals) AS Total_Medals
FROM alltime_country_rankings
GROUP BY Country
ORDER BY Total_Medals DESC;
```

   - ***Screenshot of visualisation***

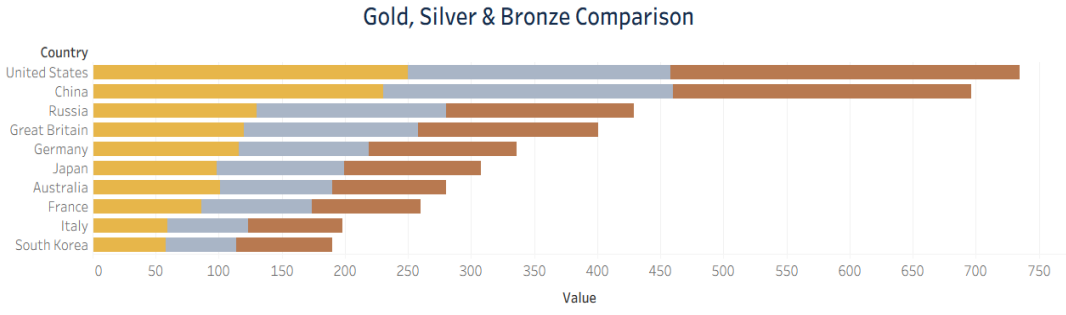


2. How do gold, silver and bronze medals compare across countries?

- ***Code***

```
SELECT
    Country,
    SUM(Total_Gold) AS Gold_Medals,
    SUM(Total_Silver) AS Silver_Medals,
    SUM(Total_Bronze) AS Bronze_Medals
FROM alltime_country_rankings
GROUP BY Country
ORDER BY Gold_Medals DESC;
```

- ***Screenshot of visualisation***

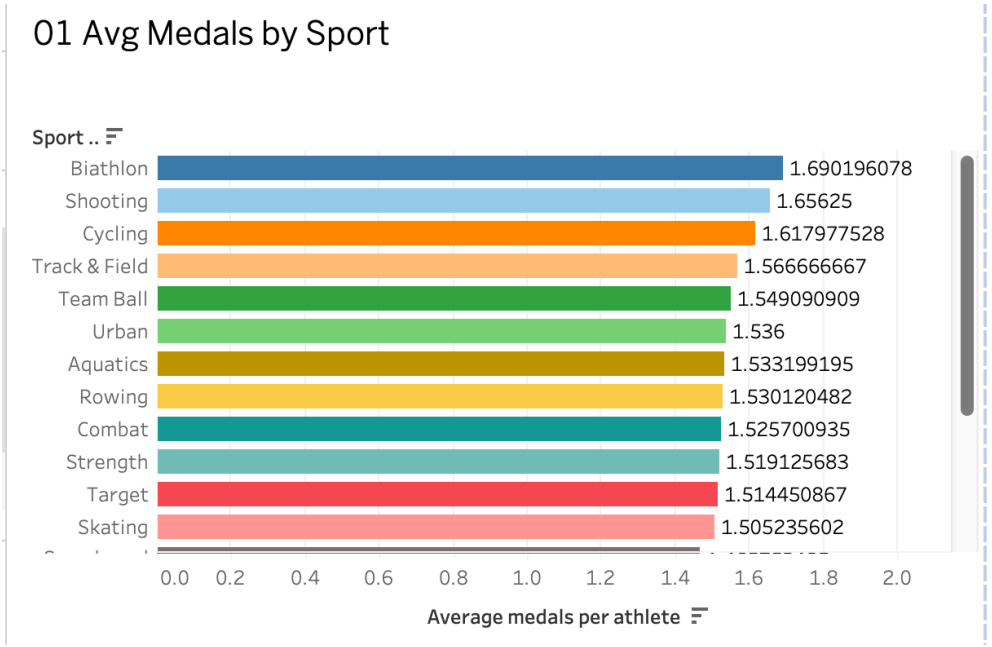


3. How have Olympic medal counts changed over the years?

- ***Code***

```
SELECT
  Year,
  SUM(Gold_Medals + Silver_Medals + Bronze_Medals)
    AS Total_Medals
FROM medal_results
GROUP BY Year
ORDER BY Year;
```

- ***Screenshot of visualisation***



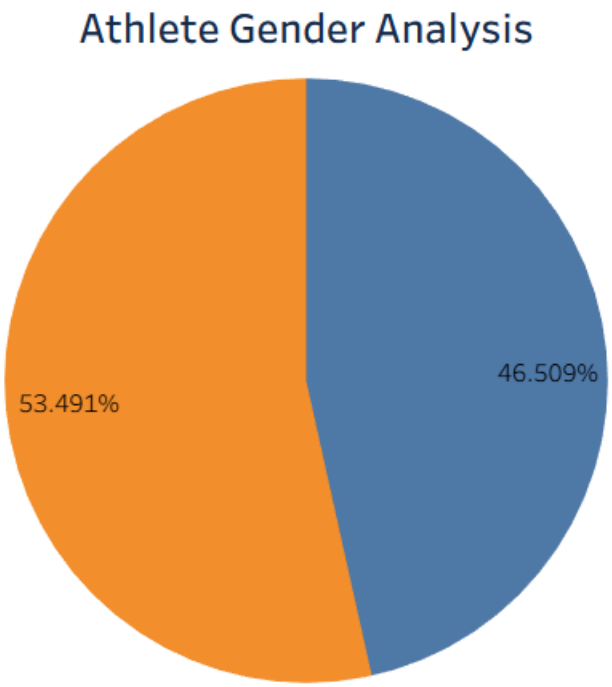
4. What is the distribution of athletes by gender?

- ***Code***

```
SELECT
  Gender,
  COUNT(DISTINCT Athlete_Id) AS Total_Athletes,
  ROUND(
    COUNT(DISTINCT Athlete_Id) * 100.0 /
    SUM(COUNT(DISTINCT Athlete_Id)) OVER (),
    2
```

```
    ) AS Percentage
FROM athletes
GROUP BY Gender;
```

- ***Screenshot of visualisation***

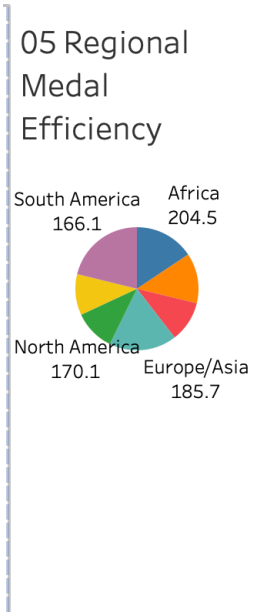


5. Which sports have the highest athlete participation?

- ***Code***

```
SELECT
    Sport,
    COUNT(DISTINCT Athlete_Id) AS Total_Athletes
FROM athletes
GROUP BY Sport
ORDER BY Total_Athletes DESC;
```

- ***Screenshot of visualisation***

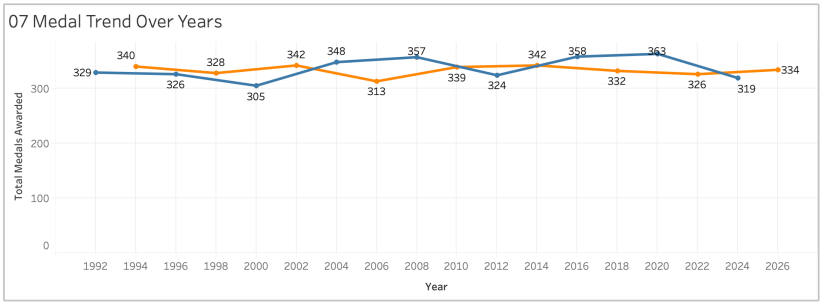


6. How does average medal efficiency vary by region?

- ***Code***

```
SELECT
    Region,
    AVG(Medal_Efficiency) AS Average_Medal_Efficiency
FROM athlete_medal_analysis
GROUP BY Region
ORDER BY Average_Medal_Efficiency DESC;
```

- ***Screenshot of visualisation***



7. How does average medal count vary across investment categories?

- ***Code***

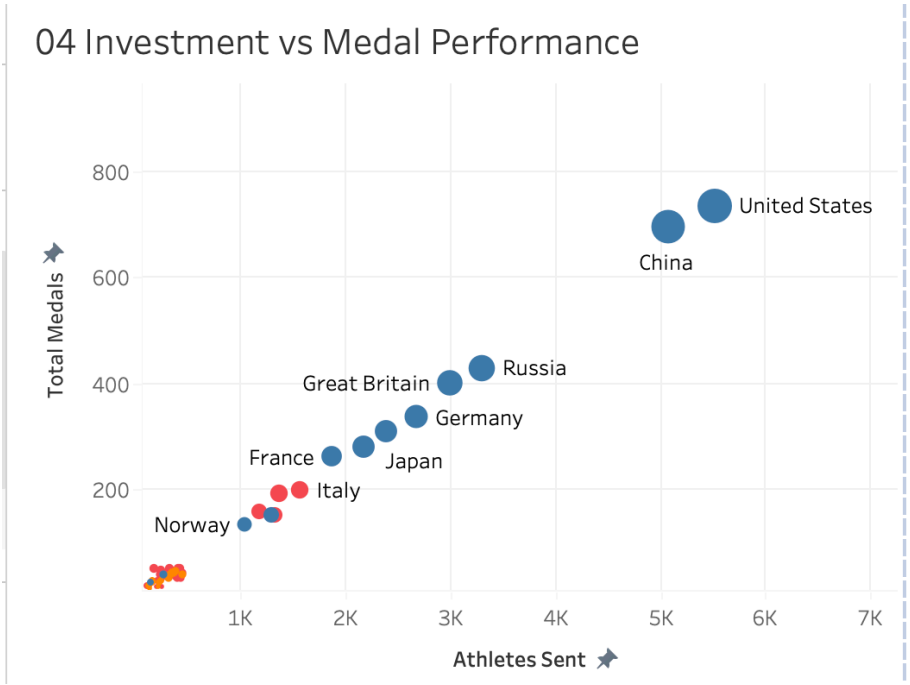
```
SELECT
    Investment_Category,
    AVG(Total_Medals) AS Average_Medal_Count
```

FROM investment_medal_analysis

GROUP BY Investment_Category

ORDER BY Average_Medal_Count DESC;

- ***Screenshot of visualisation***



8. How does medal performance vary by host country?

- ***Code***

SELECT

Host_Country,

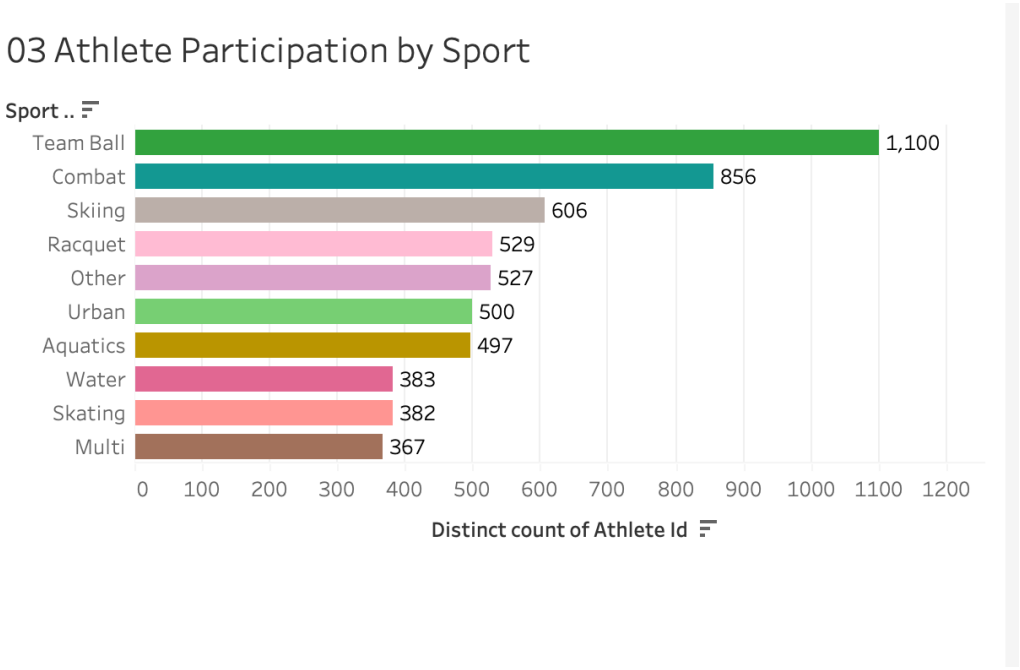
SUM(Total_Medals) AS Total_Medals

FROM medal_results

GROUP BY Host_Country

ORDER BY Total_Medals DESC;

- ***Screenshot of visualisation***


03 Athlete Participation by Sport
Sport ..
Team Ball
Combat
Skiing
Racquet
Other
Urban
Aquatics
Water
Skating
Multi
1,100
856
606
529
527
500
497
383
382
367
0
100
200
300
400
500
600
700
800
900
1000
1100
1200
Distinct count of Athlete Id